LUKUDIPLOMI — SYSTEM ARCHITECTURE

FRONTEND

React SPA | Vite Dev Server (port 5173) | HMR

React UI — **Phaser 3**

Bridge: PhaserGame.jsx

Pages

- / WelcomePage
- /login/* LoginPages
- /student/* Dashboard
- /teacher/* Dashboard
- /game PhaserGame

Phaser Scenes

- WorldMapScene
- BaseMapScene
- → 8 Continents
- ReadingScene
- ReadingState

AuthContext.jsx | Session-based Auth

React | Vite | Phaser 3

REST API

Vite proxy /api 3001

BACKEND

Node.js / Express (port 3001) | ES Modules

Controllers

Route handlers | Zod validation (zValidate)

Services

Business logic | bcrypt hashing | Error transform

Models

Knex.js queries | Data access layer

MIDDLEWARE

Error Handler

ValidationError(400) | NotFound(404) | AuthError(401)

Authentication

Passport.js (Local + Google OAuth 2.0)
express-session (server-side)
Rate limit: 5 req / 15 min

Node.js | Express | Knex.js

Knex.js

Query builder

PostgreSQL

users	Accounts & profiles
books	Book metadata
progress	Reading tracking
rewards	Achievement data
federated_credentials	OAuth links

Migrations: backend/db/migrations/

User Roles

student	default	Access game, track progress
teacher	elevated	Manage students & books
principal	reserved	Future admin features

CI / CD

GitHub Actions
Node 18/20/22 | PostgreSQL 17

Design Theme

Vintage nautical / cartographer aesthetic
Cinzel Decorative + Nunito | Finnish UI